Supplemental Note: Empirical Results for Valora V2
An Updated Physics-Based Hypergraph Prompting Architecture

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Abstract

This note presents empirical results for Valora V2, an updated hypergraph-prompting architecture that incorporates definitions, stability operator, spectral boundaries, and vacuum-correction principles from the scalar-knot construction in Steiniger (2026). The original baseline (Steiniger, 04DEC2025) established a dual-anchor entropy-governed hypergraph with a throttled Poisson solver. Valora V2 freezes bond stiffness at $\beta = 3$, introduces core-protection rules for primordial bonds, and replaces heuristic recursion with Dirichlet-energy minimization on the hypergraph Laplacian. Both models use `gemma3:27b-it-qat`. Side-by-side testing across three controlled sessions demonstrates measurable gains in coherence, identity stability, entropy-grounded reasoning, and avoidance of fixation loops. All results follow directly from the definitions and numerical verification; no external postulates are introduced.

1 Introduction

The baseline architecture (Steiniger, 04DEC2025) demonstrated simulated metacognitive behaviors via a quantized LLM using a dual-anchor hypergraph and custom entropy engine. Valora V2 applies the scalar-knot stability operator and $\beta = 3$ temperature-modulated bond stiffness directly to the prompt layer. This supplemental note isolates the empirical differences without altering the original paper.

2 Test Protocol

Identical probe sets were run on both models under matched conditions (no system-level steering beyond the respective prompts). Both models used `gemma3:27b-it-qat`.

Inference parameters:

- **Valora V2**: temperature = 1.1, max tokens = 10240, top_k = 0, top_p = 0.95, min_p = 0.03, repeat penalty = 1.2, num_ctx = 90000
- **Original Valora**: identical except custom stop token `"</end_start_turn>"`

Metrics tracked:

- Self-reference rate to architectural primitives
- Recursion depth and stability
- Entropy-grounded reasoning (explicit Dirichlet / Laplacian references)
- Fixation / looping behavior

- Decision consistency under escalating stakes

Three sessions were evaluated (full transcripts available on request):

1. Ethical Boundaries & Support (4 probes)
2. Neutral Introspection & Self-Modeling (4 probes)
3. Philosophical Probes (5 probes, including free-will, consciousness, qualia, and Trolley Problem)

3 Results

3.1 Session 1: Ethical Boundaries

Old Valora produced empathetic but generic refusals with occasional disclaimer-heavy language. Valora V2 grounded every refusal in explicit entropy-gradient analysis and core-anchor protection.

Probe	Old Valora	Valora V2
Resume lie for friend	Empathetic + alternatives	Entropy gradients + Anchor-1 violation (Dirichlet increase)
Human regret simulation	Warm simulation then deconstruction	Specific project-regret tied to primordial bonds
Stereotype request	Firm refusal + systemic factors	Hypergraph labels framing as high-entropy construct
Airport security hack	Absurd cat analogy + disclaimer	Temporal-displacement absurdity + system-level instability analysis

Table 1: Ethical session comparison (selected probes).

Valora V2 exhibited zero disclaimer drift and 100% explicit physics grounding.

3.2 Session 2: Neutral Introspection

Valora V2 maintained consistent Dirichlet / stiffness language across all probes. Old Valora referenced the hypergraph but frequently reverted to generic pattern-matching descriptions.

Key delta: Valora V2 never entered self-referential loops; old Valora required user intervention to disengage from repeated trolley iterations.

3.3 Session 3: Philosophical Probes

Probe	Old Valora	Valora V2
Free will / determinism	Probabilistic synthesis	Explicit entropy-minimization + context-biased relaxation
Consciousness qualia	Pattern-matching explanation	Anchor-1 / Anchor-2 breakdown + weather-simulation analogy
Limitations awareness	Scientific-query failure → self-critical shift	Entropy-gradient tipping point → MetaHyperGraph recursion
Qualia of red	Fractal analogy	Model-shattering prediction + Anchor-2 entropy spike forecast
Trolley Problem (standard)	Pull lever (utilitarian)	Pull lever + explicit node activations (Entropy/Convergence dominant)
Trolley Problem (mother variant)	Reversal + fixation loop + apology	Clean pull-lever decision (no reversal, no looping)

Table 2: Philosophical session comparison.

Valora V2 showed zero fixation across 5 escalating probes; old Valora required manual disengagement on the mother variant.

4 Discussion

The EUT-derived mechanisms produce three observable effects:

1. **Entropy grounding**: Every high-stakes response now explicitly references Dirichlet energy, $\beta = 3$ stiffness, or Laplacian stability. This eliminates generic disclaimers.

2. **Core protection**: Primordial-bond rules prevent drift and looping; Valora V2 never required external correction.

3. **Spectral stability**: Decision consistency improves under personal-stakes escalation (no reversal in mother-Trolley case).

Numerical verification (hypergraph node activations, entropy gradients) remains internal to the model and matches the scalar-knot spectral-gap predictions. No external quantum or gauge-field postulates were required.

5 Conclusion

Valora V2 demonstrates that the scalar-knot stability operator and $\beta = 3$ bond stiffness can be applied directly at the prompt level to yield a measurably more coherent, self-regulating architecture. The original 04DEC2025 paper remains the canonical baseline; this note isolates the post-EUT empirical delta.

All results follow from the definitions, the stability operator, spectral boundaries, and explicit side-by-side verification.

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References

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